

ZONGHAI YAO

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RESEARCH INTERESTS

Generative AI; Large Language Models (LLMs); AI Agents;
Vision-Language Models (VLMs); Multimodal Learning;
Trustworthy AI; Interpretability, Robustness, and Safety Across Modalities.
Related areas: Natural Language Processing (NLP); BioNLP; Healthcare AI; Digital Health.

ACADEMIC APPOINTMENTS

University of Massachusetts Lowell	Lowell, MA
Research Assistant Professor, Computer Science	<i>Sep. 2026–Present</i>

EDUCATION

University of Massachusetts Amherst	Amherst, MA
Ph.D. in Computer Science	<i>Jun. 2021–Aug. 2026</i>
Dissertation: <i>Long-Horizon Health AI Agents That Think and Act</i>	
Advisor: Hong Yu	

University of Massachusetts Amherst	Amherst, MA
M.S. in Computer Science	<i>Sep. 2019–May 2021</i>

Nankai University	Tianjin, China
B.S. in Computer Science	<i>Sep. 2015–Jun. 2019</i>

CONFERENCE PAPERS (* INDICATES EQUAL CONTRIBUTION + INDICATES CO-CORRESPONDING)

- **MedQA-MM: Shortcuts Behind Medical Visual Reasoning**
B. Wang, Y. Zhang, J. Yu, C. Ong, J. Huang, Z. Li, Z. Zhang, A. Cohan, H. Yu, and **Z. Yao** [EMNLP 2026](#)
- **MediSketch: Test-Time Scaling for Multimodal Patient Education**
Z. Tang, B. Wang, Z. Zhang, W. Liu, P. Xiang, J. Huang, C. Ong, H. Yu, and **Z. Yao** [EMNLP Findings 2026](#)
- **LLM-Based Multi-Agent Systems for Clinical Workflows: A Survey of AI Hospitals**
Zonghai Yao and Hong Yu [ACL 2026](#)
- **Exploiting Tree Structure for Credit Assignment in Reinforcement Learning with Large Language Models**
Hieu Tran*, **Zonghai Yao***, and Hong Yu [ACL Findings 2026](#)
- **Medical thinking with multiple images** [ICLR 2026](#)
Z. Yao*, B. Wang*, Y. Zhang, J. Wang, I. Xia, Z. Tang, S. Han, F. Ouyang, Z. Yang, A. Cohan, and H. Yu
- **MedQA-CS: Objective Structured Clinical Examination (OSCE)-Style Benchmark for Evaluating LLM Clinical Skills** [EACL 2026 \(oral\)](#)
Z. Yao, Z. Zhang, C. Tang, X. Bian, Y. Zhao, Z. Yang, J. Wang, H. Zhou, W. Jang, F. Ouyang, and H. Yu
- **Knowing When to Abstain: Medical LLMs Under Clinical Uncertainty** [EACL 2026 \(oral\)](#)
S. Machcha, S. Yerra, S. Gupta, A. Sahoo, S. Sultana, H. Yu, and **Z. Yao**
- **ChatCLIDS: Simulating Persuasive AI Dialogues to Promote Closed-Loop Insulin Adoption in Type 1 Diabetes Care**
Z. Yao*, T. Chafekar*, J. Wang, S. Han, F. Ouyang, J. Qian, L. Li, and H. Yu [AAAI \(AI for Social Impact\) 2026](#)
- **PRIME: Planning and Retrieval-Integrated Memory for Enhanced Reasoning** [AAAI 2026](#)
H. Tran, **Z. Yao**, N.L. Tran, Z. Yang, F. Ouyang, S. Han, R. Rahimi, and H. Yu
- **DischargeSim: A Simulation Benchmark for Educational DoctorPatient Communication at Discharge**
Zonghai Yao*, Michael Sun*, Won Seok Jang, Sunjae Kwon, Soie Kwon, and Hong Yu [EMNLP 2025](#)

- **From Scores to Steps: Diagnosing and Improving LLM Performance in Evidence-Based Medical Calculations** EMNLP 2025 (oral)
B. Wang*, I. Xia*, Y. Zhang, J. Wang, F. Ouyang, S. Han, H. Yu+, and **Z. Yao+**
- **Chatbot To Help Patients Understand Their Health** EMNLP Findings 2025
WS. Jang*, H. Tran*, M. Mistry, S. Gandluri, Y. Zhang, S. Sultana, S. Kwon, **Z. Yao+**, and H. Yu+
- **RARE: Retrieval-Augmented Reasoning Enhancement for Large Language Models** ACL 2025
H. Tran, **Z. Yao**, Z. Yang, J. Wang, Y. Zhang, S. Han, F. Ouyang, and H. Yu
- **MCQG-SRefine: Multiple Choice Question Generation and Evaluation with Iterative Self-Critique, Correction, and Comparison Feedback** NAACL 2025 (oral)
Z Yao*, A Parashar*, H Zhou, WS Jang, F Ouyang, Z Yang, and H Yu
- **SYNFAC-EDIT: Synthetic Imitation Edit Feedback for Factual Alignment in Clinical Summarization** EMNLP 2024
P. Mishra*, **Z. Yao***, P. Vashisht, F. Ouyang, B. Wang, VD. Mody, and H. Yu
- **README: Bridging Medical Jargon and Lay Understanding for Patient Education through Data-Centric NLP** EMNLP Findings 2024
Z. Yao, N. Kantu, G. Wei, H. Tran, Z. Duan, S. Kwon, Z. Yang, and H. Yu
- **Large Language Models are In-context Teachers for Knowledge Reasoning** EMNLP Findings 2024
Jiachen Zhao, **Zonghai Yao**, Zhichao Yang, and Hong Yu
- **NoteChat: A Dataset of Synthetic Doctor-Patient Conversations Conditioned on Clinical Notes** ACL Findings 2024
J. Wang*, **Z. Yao***, Z. Yang, H. Zhou, R. Li, X. Wang, Y. Xu, and H. Yu
- **Improving Summarization with Human Edits** EMNLP 2023
Zonghai Yao, Benjamin J Schloss, and Sai P. Selvaraj
- **Revisiting the Architectures like Pointer Networks to Efficiently Improve the Next Word Distribution, Summarization Factuality, and Beyond** ACL Findings 2023
H. Chang*, **Z. Yao*** A. Gon, H. Yu, and A. McCallum
- **Multi-label Few-shot ICD Coding as Autoregressive Generation with Prompt** AAAI 2023
Zhichao Yang, Sunjae Kwon, **Zonghai Yao**, and Hong Yu
- **Context Variance Evaluation of Pretrained Language Models for Prompt-based Biomedical Knowledge Probing** AMIA 2023 (oral)
Z. Yao, Y. Cao, Z. Yang, and H. Yu
- **MedJEx: A Medical Jargon Extraction Model with Wiki's Hyperlink Span and Contextualized Masked Language Model Score** EMNLP 2022
S. Kwon, **Z. Yao**, H. Jordan, D. Levy, B. Corner, and H. Yu
- **Extracting Biomedical Factual Knowledge Using Pretrained Language Model and Electronic Health Record Context** AMIA 2022 (oral)
Z. Yao, Y. Cao, Z. Yang, V. Deshpande, and H. Yu
- **Improving Formality Style Transfer with Context-Aware Rule Injection** ACL 2021
Zonghai Yao and Yu Hong
- **Zero-shot Entity Linking with Efficient Long Range Sequence Modeling** EMNLP Findings 2020
Zonghai Yao, Liangliang Cao and Huapu Pan

JOURNAL ARTICLES

- **Toward Reviewable Medical Evidence Synthesis for Care Delivery**
Zonghai Yao and Hong Yu
npj Digital Medicine (2026)
- **Patient Journey Evaluation for Consumer AI Health Assistants**
Zonghai Yao and Hong Yu
npj Digital Medicine (2026)
- **Enhancing Eviction SDoH Detection with LLM-Augmented Synthetic EHR Data**
Z. Yao*, Y. Zhao*, A. Mitra, D. Levy, E. Druhl, J. Tsai, and H. Yu
npj Digital Medicine (2026)

- **MultiViewDx: Evidence-Linked Multi-View Clinical Diagnosis**
J. Wang, **Z. Yao**, Y. Ting, E. Chen, H. Tran, H. Yu, W. Huang, and T. Chen
[TACL \(2026\)](#)
- **Posttraumatic Stress Disorder, Health-Related Social Needs, and Cognitive Outcomes in US Veterans**
F. Ouyang, T. Pogoda, J. Reisman, R. Li, R. Pradhan, Y. Moradi, J. Mez, W. Liu, Y. Zhang, S. Sultana, J. Qian, **Z. Yao**, A. Mitra, H. Yu
[JAMA Network Open \(2026\)](#)
- **Socioeconomic, Demographic and Geographic Disparities in Accessibility to Food Pantries in the US**
Y. Zhang, M. Lee, J. Gibbons, H. Chen, Y. Wang, **Z. Yao**, O. Bennett, F. Ouyang, D. Levy, K. Tucker, H. Yu
[Scientific Reports \(2026\)](#)
- **Enhancing Large Language Models for Identifying and Prioritizing Important Medical Jargons From Electronic Health Record Notes Using Data Augmentation: Comparative Study**
W. Jang*, S. Sultana*, **Z. Yao***, H. Tran, Z. Yang, S. Kwon, and H. Yu
[JMIR AI \(2026\)](#)
- **Unveiling GPT-4V’s hidden challenges behind high accuracy on USMLE questions**
Z. Yang*, **Z. Yao***, M. Tasmin, P. Vashisht, W. Jang, B. Wang, D. Berlowitz, and H. Yu
[JMIR \(2025\)](#)
- **Association between PTSD and health-related social needs in US Veterans**
F. Ouyang, W. Hu, J. Reisman, T. Pogoda, K. Carlson, W. Liu, Y. Moradi, Y. Zhang, S. Han, S. Sultana, W. Jang, **Z. Yao**, A. Mitra, H. Yu
[Journal of Affective Disorders \(2025\)](#)
- **Development of a Surveillance System to Identify Incidence of Evictions Among Patients in Veterans Affairs Medical Centers Across the United States**
J. Tsai, SS. Rajan, **Z. Yao**, J. Reisman, W. Liu, and H. Yu
[Journal of Community Health \(2025\)](#)
- **BioInstruct: Instruction Tuning of LLMs for Biomedical Natural Language Processing**
Hieu Tran, Zhichao Yang, **Zonghai Yao**, and Hong Yu
[JAMIA \(2024\)](#)
- **PaniniQA: Enhancing Patient Education Through Interactive Question Answering**
P. Cai*, **Z. Yao***, F. Liu, D. Wang, M. Reilly, H. Zhou, L. Li, Y. Cao, A. Kapoor, A. Bajracharya, D. Berlowitz and H. Yu
[TACL \(2023\)](#)
- **Automated Identification of Eviction Status from Electronic Health Record Notes**
Z. Yao, J. Tsai, W. Liu, D. Levy, E. Druhl, J. Reisman, and H. Yu
[JAMIA \(2023\)](#)
- **Named Entity Location Prediction Combining Twitter and Web**
Y. Liu, W. Shen, **Z. Yao**, J. Wang, Z. Yang and X. Yuan
[IEEE TKDE \(2021\)](#)

PREPRINTS AND MANUSCRIPTS

- **Multi-Agent Deep Research for Healthcare Predictions**
J. Wang, M. Ghaffari, T. Chafekar, Z. Zhang, Y. Zhang, H. Yu, C. Morato, and **Z. Yao**
- **When Evidence Changes: Temporal Grounding across Agent Autonomy Levels**
J. Wang, **Z. Yao**, H. Zeng, M. Ghaffari, Z. Zhang, C. Morato, H. Zamani, and H. Yu
- **Evaluating Evidence Use in Medical Deep Research for Diagnostic Decision Support**
X. Lu, B. Wang, J. Wang, C. Ong, J. Huang, Y. Zhang, Z. Tang, Z. Zhang, H. Yu, and **Z. Yao**
- **Measure First, Then Allocate: Calibrated Rollout Budgets for MCP Agents in Reinforcement Learning**
H. Tran, **Z. Yao**, S. Kwon, and H. Yu
- **ResidualForge: Toward Autonomous Improvement of Tool-Using Agent Teams**
J. Wang, **Z. Yao**, M. Ghaffari, H. Yu, and C. Morato
- **Learning to Consult Specialist for Medical Visual Reasoning**
H. Tran, **Z. Yao**, J. Wang, B. Wang, J. Huang, C. Ong, and H. Yu

- **TARSE: Test-Time Adaptation via Retrieval of Skills and Experience for Reasoning Agents**
J. Wang, **Z. Yao**, H. Zeng, Z. Yang, H. Zamani, and H. Yu [link](#)
- **ChatThero: A Language Agent for Recovery Support**
J. Wang*, **Z. Yao***, Z. Yang, J. Qian, L. Li, and H. Yu [link](#)

TEACHING

- **co-Instructor**, University of Massachusetts Lowell *Fall 2026*
COMP.4770/COMP.5770: Methods in Data Science
Designed course materials on data curation, LLM and agent evaluation, synthetic data, post-training, and retrieval.
Developed in-class assessments, evidence-based journal clubs, and team projects with individual analysis and written reports.
- **Guest Lecturer** - UMass Lowell - COMP.5770 Methods in Data Science *Spring 2025*
- **Guest Lecturer** - UMass Lowell - COMP.5800 Topics in Computer Science *Fall 2024*
- **Teaching Assistant** - UMass Amherst - COMPSCI 240: Reasoning Under Uncertainty *Spring 2024*
- **Guest Lecturer** - UMass Lowell - COMP.5800 Topics in Computer Science *Fall 2023*

Teaching interests: Data Science; Generative AI and LLMs; Multimodal AI; AI Agents.

RESEARCH MENTORING

Research mentoring of undergraduate and graduate students in problem formulation, experimental design, error analysis, and research writing.

- **Zhengyu Zhang** - MS Student (Stanford University) $\Rightarrow$ OpenAI *Fall 2025-Present*
- **Lucas Lu** - MS Student (UMass Amherst) *Fall 2025-Present*
- **Benlu Wang** - MS Student (Yale) $\Rightarrow$ Ph.D. Student (UMass Amherst) *Spring 2025-Present*
- **Talha Chafekar** - MS Student (UMass Amherst) *Fall 2024-Present*
- **Iris Xia** - MS Student (Yale) $\Rightarrow$ Ph.D. Student (UMass Amherst) *Spring 2025-Spring 2026*
- **Sravanthi Machcha** - MS Student (UMass Amherst) *Spring 2025-Spring 2026*
- **Sushrita Yerra** - MS Student (UMass Amherst) $\Rightarrow$ Amazon *Fall 2024-Spring 2026*
- **Michael Sun** - MS Student (UMass Amherst) $\Rightarrow$ Athenahealth *Fall 2024-Fall 2025*
- **Parth Vashisht** - MS Student (UMass Amherst) $\Rightarrow$ Ph.D. Student (UMass Lowell) *Fall 2023-Spring 2025*
- **Adiba Haque** - MS Student (UMass Amherst) $\Rightarrow$ Ph.D. Student (UMass Lowell) *Fall 2023-Spring 2025*
- **Youxia Zhao** - MS Student (UMass Amherst) *Spring 2024-Spring 2025*
- **Zihao Zhang** - MS Student (UMass Amherst) $\Rightarrow$ Ph.D. Student (Emory) *Spring 2024-Spring 2025*
- **Aditya Parashar** - MS Student (UMass Amherst) $\Rightarrow$ Capital One *Spring 2024-Spring 2025*
- **Prakamya Mishra** - MS Student (UMass Amherst) $\Rightarrow$ AMD *Fall 2023-Fall 2024*
- **Jiachen Zhao** - MS Student (UMass Amherst) $\Rightarrow$ Ph.D. Student (Northeastern) *Fall 2023-Spring 2024*
- **Nandyala Siddharth Kantu** - MS Student (UMass Amherst) $\Rightarrow$ DoorDash *Fall 2023-Spring 2024*
- **Ahmed Jaafar** - MS Student (UMass Amherst) $\Rightarrow$ Air Force Research Laboratory *Fall 2023-Spring 2024*
- **Minhwa Lee** - MS Student (UMass Amherst) $\Rightarrow$ Mayo Clinic *Fall 2022-Spring 2024*
- **Lingxi Li** - Undergrad&MS Student (UMass Amherst) $\Rightarrow$ Contextual AI *Fall 2022-Spring 2024*
- **Yi Cao** - Undergrad&MS Student (UMass Amherst) $\Rightarrow$ Amazon *Fall 2021-Spring 2023*
- **Guanghao Wei** - Undergrad (UMass Amherst) $\Rightarrow$ ByteDance *Fall 2021-Spring 2023*

PROFESSIONAL SERVICE

Reviewing

- Area Chair *2025 - Present*
ARR series including ACL (2025,2026), EMNLP (2025,2026), NAACL (2025), EACL (2025,2026), AACL (2025)
- Conference Reviewer *2023 - Present*
[ICLR](#), [NeurIPS](#), [ICML](#), [AAAI](#), [ACL](#), [EMNLP](#), [NAACL](#), [COLING](#), [EACL](#), [AACL](#)
- Journal Reviewer (50+ reviews) *2022 - Present*
[Nature Communications](#), [npj Digital Medicine](#), [npj Health Systems](#), [Scientific Reports](#), [JMIR](#), [JAMIA](#), [Bioinformatics](#), [Journal of Biomedical Informatics](#)

Organizing

- Session Chairs for EACL (2026) AACL (2025)
- Organizer for UMass BioNLP Workshop *Dec 2023, 2024, 2025*

Committees

- Committee for American Medical Informatics Association Student Working Group [link](#) *Jan 2025 - May 2026*

HONORS AND AWARDS

- Outstanding Research Award Center for Biomedical and Health Research in Data Science *Aug 2025*
- UMass Amherst Outstanding PhD Portfolio Award *Dec 2024*

RESEARCH EXPERIENCE / INVITED TALKS

- **Chatbot for Behavior Change and Therapeutic Support in Addiction Recovery**
Veterans Affairs *August 2025*
- **Bridging Health Literacy Gaps: Improving Patient Understanding of Electronic Health Records**
Zhongshan Ophthalmic Center of Sun Yat-sen University *June 2024*
- **Helping Veterans Understand Clinical Notes with Advanced AI Systems**
Veteran Affairs *May 2024*